Variability and reliability of hBN-encapsulated 2D FETs

Fernando de los Santos-Prieto,
Theresia Knobloch, Tibor Grasser

Institute for Microelectronics, TU Wien

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Outline

- Introduction
- Motivation and Device Fabrication
- Performance and Variability
  - Device Performance
  - Device Variability
- Reliability
  - Hysteresis
  - Bias Temperature Instability

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Introduction

Demand of new gate stacks for 2D FETs

- The transition from silicon to 2D materials at the channel demands new and highly optimized gate stacks and insulators.
- The construction of Van der Waals (VdW) heterostructures using 2D insulators such as hexagonal boron nitride (hBN) is considered one of the most promising alternatives.

Ongoing research and challenges

- Although its high leakage current prevents hBN from being used as the sole insulator in the stack[a], encapsulation with hBN monolayers has gained attention as a performance and reliability enhancer for 2D FETs.

[a] Knobloch et al. 2021.

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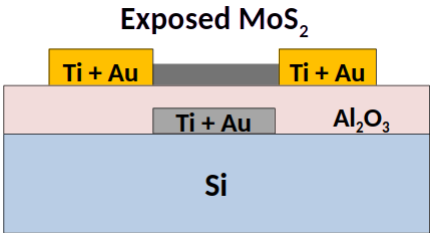
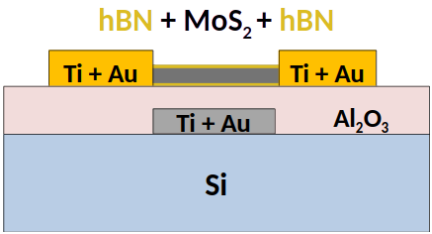
Motivation and Device Fabrication

Limited Systematic Evaluation of hBN-encapsulated FETs

- Due to the immaturity of the technology and the lack of standardized measurement schemes, these novel gate stacks involving hBN monolayers have not yet been systematically characterized.
- To enable such comprehensive evaluation of hBN encapsulation, structurally similar transistors with and without hBN were fabricated.





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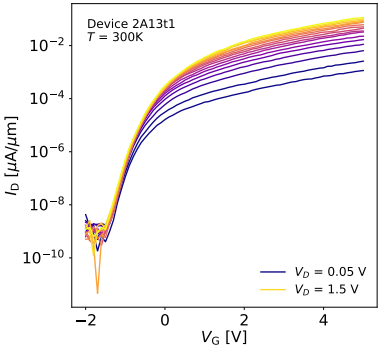
Device Performance and Variability

Performance

- hBN-encapsulated transistors operate both in vacuum and under ambient conditions, whereas non-encapsulated devices require vacuum and high-temperature annealing prior to operation.
- In terms of performance metrics, encapsulated devices exhibit a non-ideal subthreshold slope of approximately 200 mV/dec and a maximum contact-limited current of about 0.1 μA/μm.



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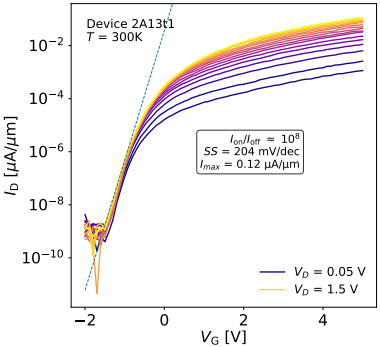
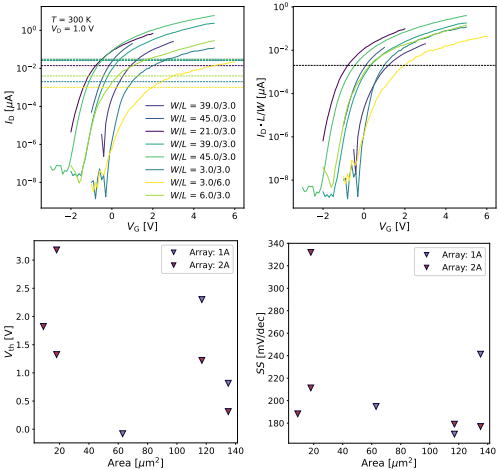


Device Performance and Variability

Variability

- Different aspect-ratio (W/L) transistors were fabricated for the encapsulated devices,ranging from 45/3 to 3/6 (in μm).
- hBN-encapsulated devices exhibit high V_{th} variability with only a weak dependence on device area.
- The subthreshold slope exhibits low variability, with all values within the range of 180–240 mV/dec, except for the long-channel (3/6) device.



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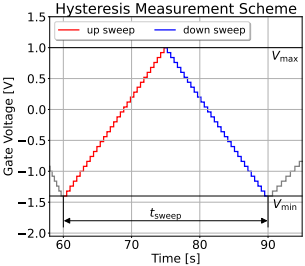
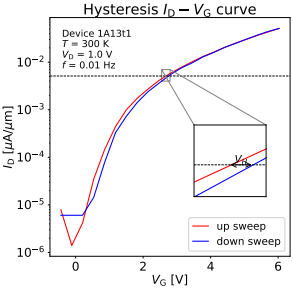
Hysteresis

- Hysteresis measurements (i.e., forward and reverse sweeps of the I_D-V_G characteristics) are the most widely used method to assess the reliability of 2D FETs.
- However, hysteresis results are often reported without specifying key measurement parameters, such as the applied voltage range or the measurement time (frequency).

Standarization of hysteresis

- Voltage ranges chosen to normalize across samples of different thicknesses[1].
- Varying sweep frequency to capture time-dependent reliability effects.



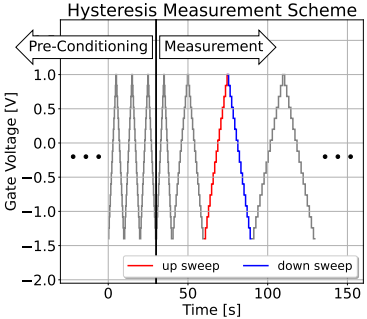


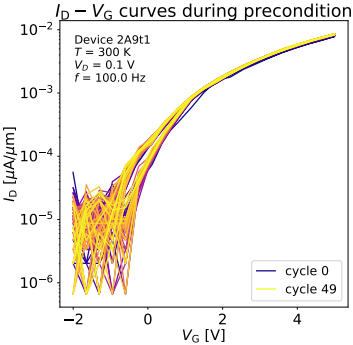
[1] Karl et al. 2025.

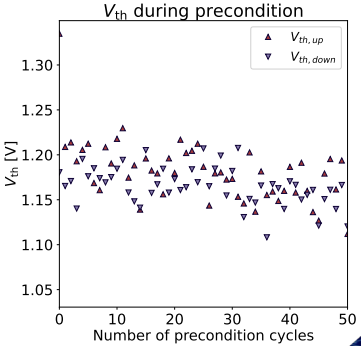
Hysteresis measurements

Precondition

- It has been experimentally observed that the initial hysteresis measurements may vary due to the undefined initial state of the device.
- However, it has been shown that the devices reach a cyclostationary response after a few cycles. To exploit this, a precondition scheme can be implemented.







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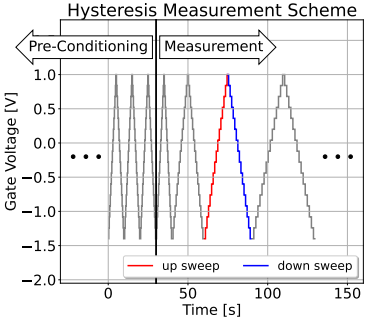
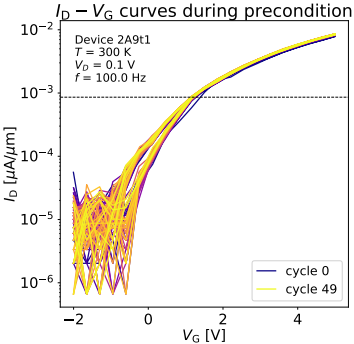
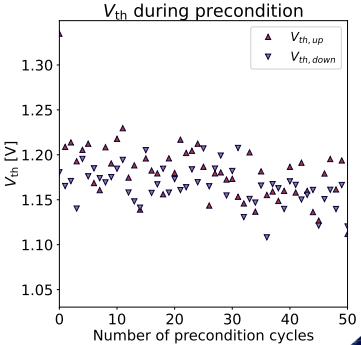


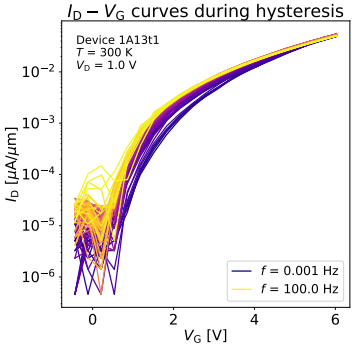




Hysteresis measurements

A standarized hysteresis measurement

1. Appropiate voltage range according to normalized overdrive electric field, $E_{\text{od}} = \frac{V_{\text{G}} - V_{\text{th}}}{\text{EOT}}$.



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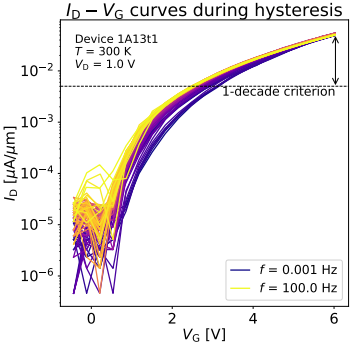

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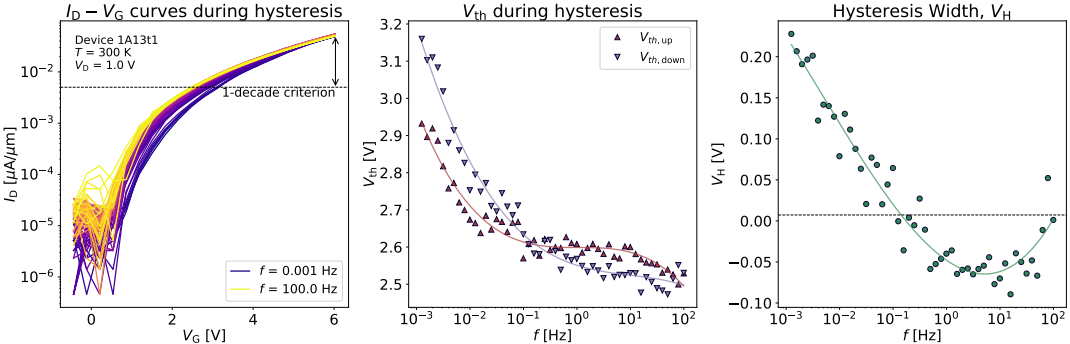

Hysteresis measurements

Importance of the drain voltage

- Ideally, V_D should be as low as possible when probing the reliability of the gate stack but, often, a higher V_D is employed to obtain accurate current levels.
- Explore using different V_D values in the hysteresis measurements as a sanity check before drawing conclusions about the stack.

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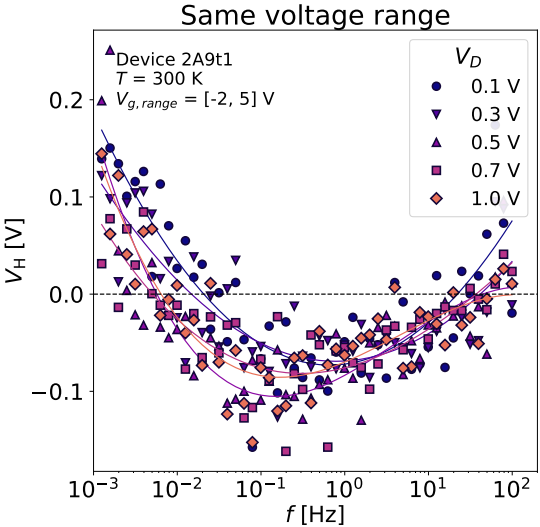

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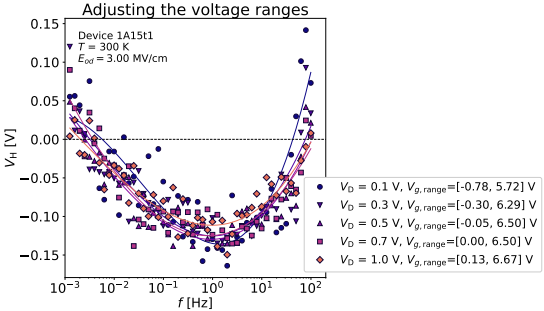
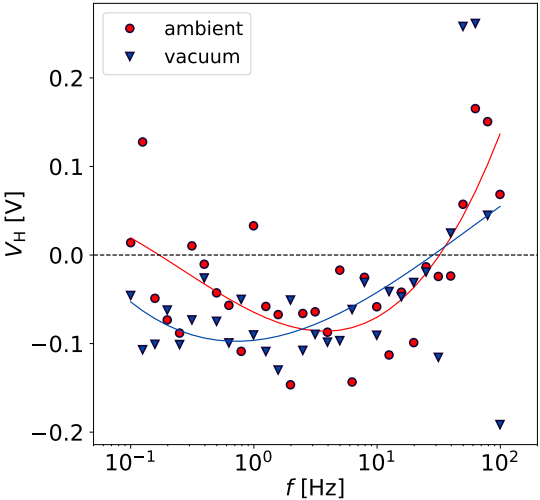


Hysteresis measurements

Ambient vs vacuum

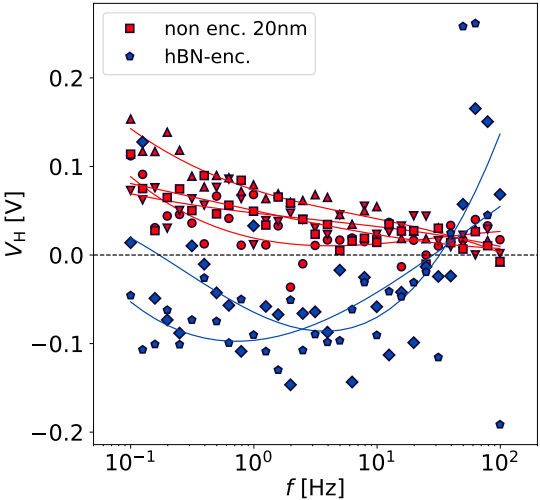
- In contrast to exposed MoS_2 devices, the encapsulated devices operate similarly regardless of the ambient conditions.
- No significant difference was observed in the hysteresis behavior when comparing vacuum and ambient conditions.



Hysteresis measurements

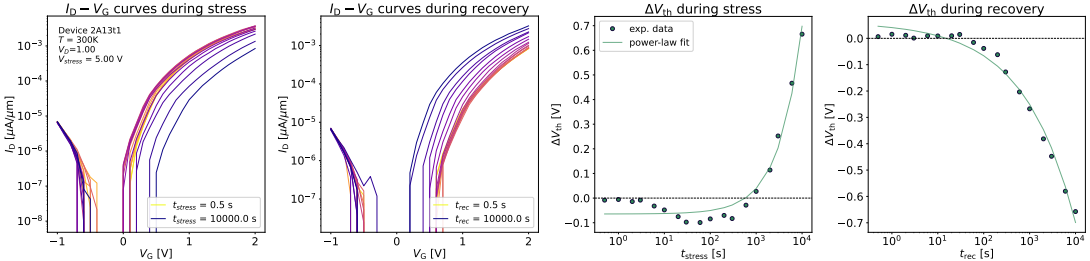
hBN-encapsulated vs exposed MoS_2

- Only after prolonged high-temperature annealing were the exposed MoS_2 devices sufficiently stable to perform hysteresis measurements.
- Both devices exhibit similar absolute hysteresis widths, but with opposite signs.



Bias Temperature Instability

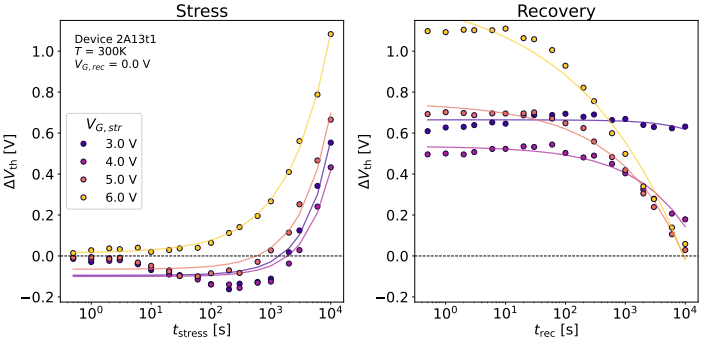
- Complementary to hysteresis measurements, bias temperature instability (BTI) experiments can also be used to assess the reliability of the gate stack.
- In contrast to hysteresis, BTI measurements involve constant biases (with the exception of the quick I_D-V_G sweeps) at the gate stack, reducing the noise in the measurements.
- BTI measurements can be performed either on-the-fly (OTF), where both the stress and recovery periods are measured, or with the measure-stress-measure (MSM) scheme, where only recovery data is collected.



Bias Temperature Instability

OTF measuremsents with different stress voltages

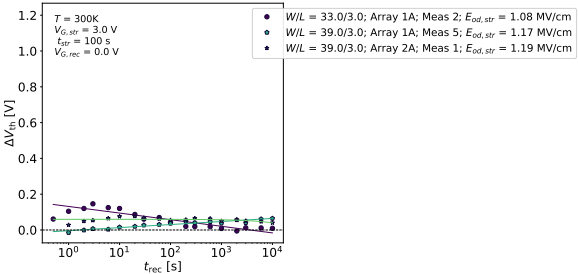
- Exploring different stress voltages provide insights into the underlying degradation mechanisms.
- In the hBN-encapsulated devices, it is important to notice the stability of the devices for short periods of times for both stress and recovery.



Bias Temperature Instability

MSM measurements for different aspect-ratios

- MSM measurements have the advantage of not disturbing the stress period and, thus, providing a more accurate degradation values just after stress.
- This type of measurement have been performed for different aspect-ratio devices and for different stress voltages, showing high variability in the initial degradation values.



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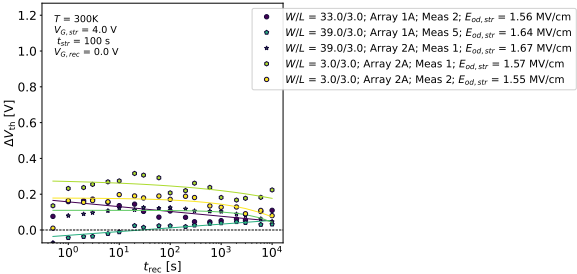

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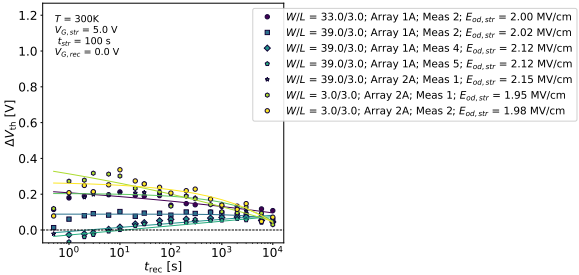

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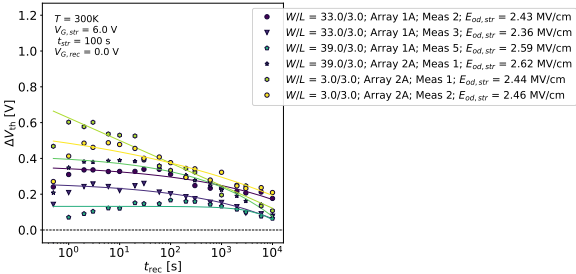

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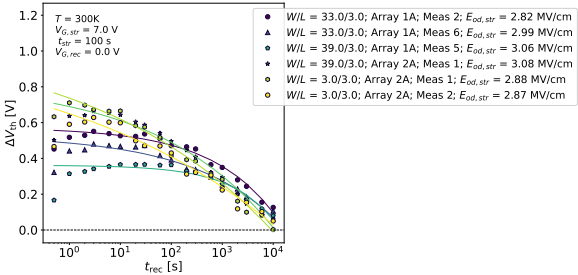

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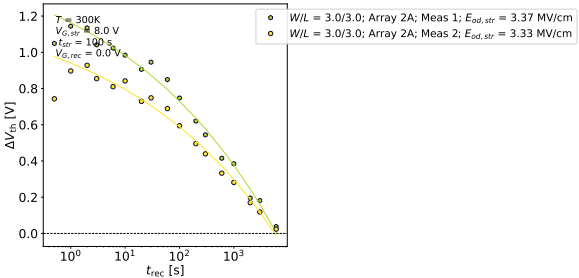
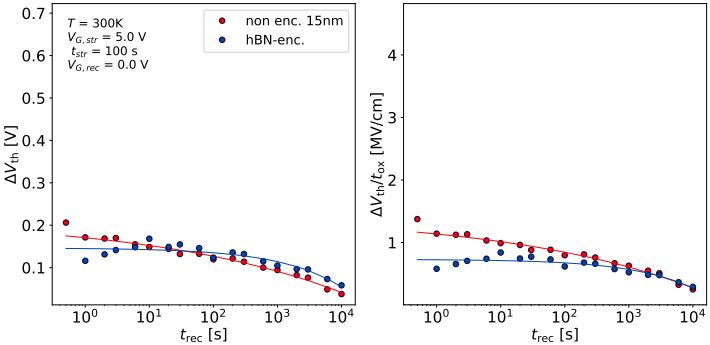


Bias Temperature Instability

Comparison hBN-encapsulated vs exposed MoS_2

- MSM data was also collected for the exposed MoS_2 devices with an oxide thickness of 15-nm. The average degradation values of both stacks are comparable for the same stress voltage.
- However, a normalization of the degradation values by the oxide thickness (i.e., $\Delta V_{th}/t_{ox}$) reveals a higher degradation for the exposed MoS_2 devices.



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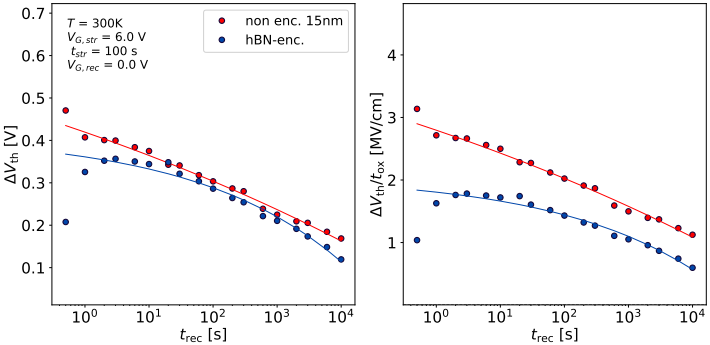

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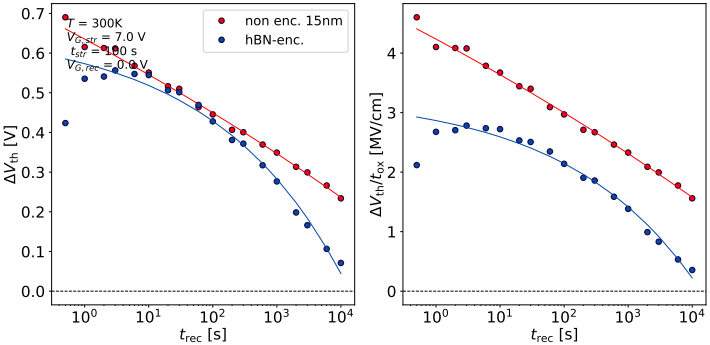

Conclusions and Future Work

Conclusions

- hBN encapsulation enables the operation of MoS_2 FETs under ambient conditions, but the devices still exhibit non-ideal performance metrics.
- Hysteresis measurements show similar absolute hysteresis widths for both exposed and encapsulated devices, but with opposite signs.
- BTI measurements reveal high variability in the initial degradation values for encapsulated devices, with no clear dependence on the device aspect-ratio. However, when normalizing the degradation by the oxide thickness, the exposed MoS_2 devices exhibit higher degradation compared to the encapsulated ones.

Future work

- Performing additional measurements such as CV measurements to further understand the the interface and the degradation mechanisms in the hBN-encapsulated devices.

Fernando de los Santos-Prieto, Theresia Knobloch, Tibor Grasser, delossa@iue.tuwien.ac.at